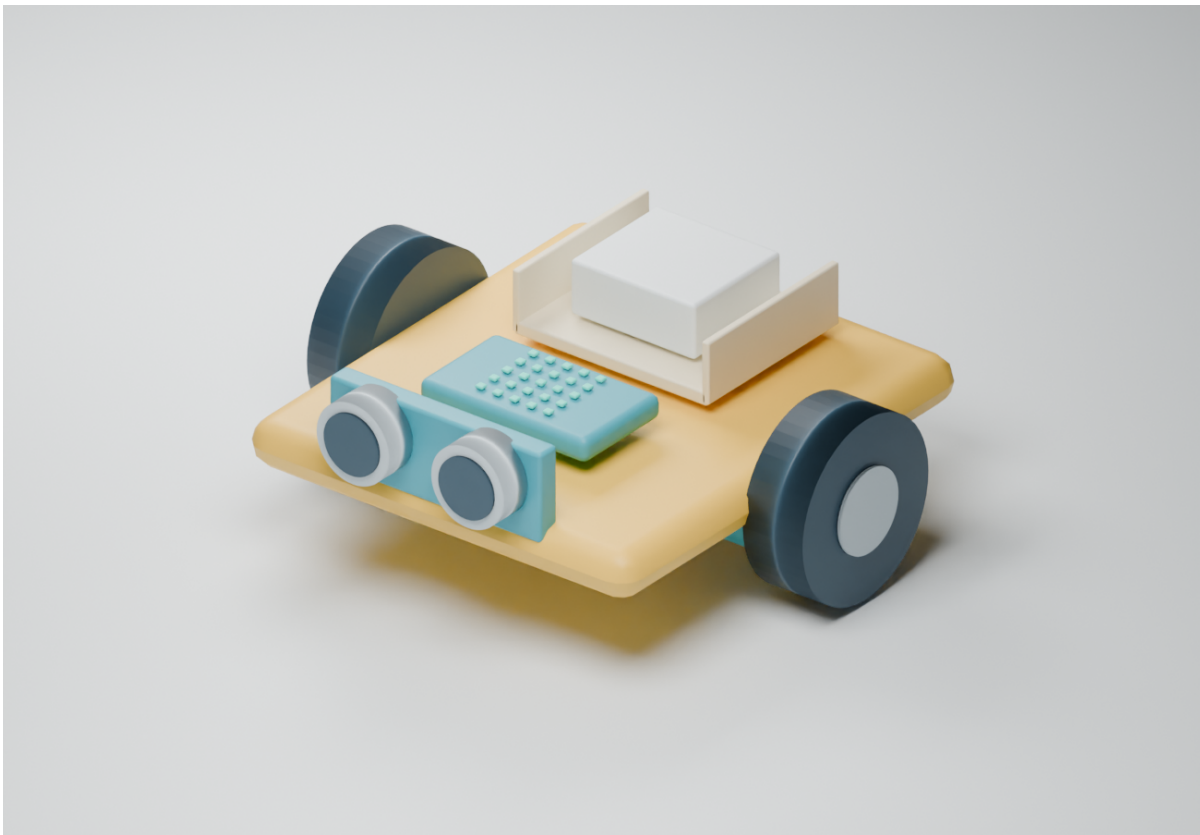


Build your first delivery rover

Teacher guide • setup, facilitation, answer key and rubric

Grades 6-8 • 8 modules • 16 sessions of 45 minutes

BBC micro:bit V2 + Kitronik :MOVE Motor (5683)



Conceptual Blender illustration. Use the manufacturer guide for actual assembly.

Predict. Build. Test. Explain. Improve.

Build a robot that can deliver a small paper parcel, detect an obstacle, and stop at its destination. Work like an engineer: predict, build, test, explain, and improve.

Set up for a dependable lesson

Scope and readiness

This is a complete teaching draft for paired learners in grades 6-8. It assumes basic computer navigation, but no previous robotics coding. Reading and multiplication support may be needed. Complete the teacher hardware preflight before classroom use; these programs have not been tested on a physical robot in this project.

Materials and preparation

For each pair: one micro:bit V2, one Kitronik :MOVE Motor (5683), four AA batteries of the type recommended by Kitronik, a data-capable micro-USB cable, and a computer with MakeCode access. The micro:bit is sold separately from some robot kits.

Shared supplies: ruler or tape measure, masking tape, large white paper, dark tape, cardboard, paper, reusable adhesive, scissors used under school procedures, and an upright cardboard obstacle at least as wide as the robot. Use a flat floor test lane, away from feet and stairs.

Teacher preflight: assemble the supplied kit using its manufacturer guide. Fit the supplied parts as directed; this course adds a removable paper cargo cradle rather than rebuilding the electronics or chassis. Insert the micro:bit only with robot power off and in the manufacturer's stated orientation.

At makecode.microbit.org create a project. Open Extensions, search "move motor", and choose Kitronik's kitronik-move-motor extension. For the supplied reference program, switch to JavaScript, replace the starter text with the lesson's .ts file, then switch to Blocks if desired. Set the indicated tuning values before downloading.

With robot battery power off, transfer the program by USB using MakeCode's device connection or copy its downloaded .hex file onto the MICROBIT drive. Disconnect the USB cable before floor driving. Use the robot's specified AA supply for driving; do not connect a separate micro:bit battery pack at the same time.

Run the classroom

Before learners arrive

For every pair: check assembly, fresh suitable batteries, current program transfer, forward direction, B stop, power switch, sensor at 10/20/30 cm, and a low-speed stopping trial. Mark separate floor lanes. Reserve one known-working robot for diagnosis. Label each kit and keep its calibration log with it.

Safety and supervision

Start on a clear floor with an adult ready to switch off the robot. Button A starts a trial and button B requests a software stop. The physical power switch is the immediate stop if code or controls fail. Reference programs also have a short time limit. Test each program on the actual kit before class; wheel motion and stopping distances require local calibration.

Students rotate driver/programmer and observer/data-recorder roles after each trial. Keep hair, fingers and loose clothing away from wheels. Switch power off before adjusting the robot; keep liquids and metal objects away from the board. Follow the school's and manufacturers' safety instructions.

Session rhythm

Session A: launch and predict (5 min), explain/model (10), build and program (20), first test (5), save and tidy (5).

Session B: retrieval (5 min), fair tests and revisions (25), explanation or peer review (10), exit ticket and tidy (5).

Support and extension

Provide printed arrow cards and preloaded programs. Read prompts aloud or use narration and transcripts. Permit diagrams plus an oral explanation instead of extended writing. Rotate roles so every learner programs and explains. For advanced learners, require a new constraint and an evidence-based test, rather than simply increasing speed.

If hardware is unavailable

Use the grid simulator for sequences, decisions and evidence logs, plus paper cradle prototypes. Label these as simulated or paper results. Schedule the physical build, measurement and stop checks when kits are available; simulator completion does not establish physical robotics proficiency.

Assessment evidence

Collect an individual exit explanation each module, a team trial log, and the final handover. Grade engineering reasoning and honest evidence as well as whether the rover completes a run.

Course map

Mission	Physical deliverable	Concept
1. Sense, decide, respond	Build a rover that responds to its operator	Input / event / output
2. Motor checkout	Make a controlled, repeatable drive	Drive command and measurement
3. Design a cargo cradle	Carry a paper parcel without blocking the robot	Constraints and attachment design
4. Give the rover reliable senses	Build and calibrate a parking-distance display	Sensor calibration and uncertainty
5. Code a delivery route	Drive to a marked delivery bay	Sequence and open-loop control
6. Stop before an obstacle	Build a cautious autonomous approach	Conditional and latched stop
7. Plan the delivery mission	Deliver a paper parcel using a tested approach	Integration and acceptance tests
8. Showcase with evidence	Prove, explain and hand over your robot	Handover and evidence

Scheduling options

Standard: two sessions per week for eight weeks. Intensive: four sessions per week for four weeks. Keep the repeated trials and reflections in either schedule. Allow an extra setup session if the kits arrive unassembled.

Sense, decide, respond

Learning evidence

Identify an input, a programmed decision and an output, then demonstrate a predictable response in three trials.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

A shows the chosen ready icon, B shows the stop icon, and wheels remain still in all three trials.

Exit question and answer key

A remotely operated exploration robot is still a robot. What extra capability would make this rover autonomous?

It would read an environmental sensor and use a programmed rule to choose an action without a person choosing every movement.

Misconception to address

“Think” means execute programmed rules; it does not imply that a micro:bit understands or has intentions.

Troubleshooting

If the display stays unchanged, check that the new program transferred and that you are pressing the correct button. If a motor moves, switch off and load the correct lesson program.

Stretch task

Add a distinct icon for A+B. Explain why a recognizable stop state helps another operator.

Motor checkout

Learning evidence

Run a bounded forward drive, demonstrate stop, and identify two sources of variation.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Three automatic stops and a demonstrated B stop, with travel distance and variation recorded.

Exit question and answer key

Why might speed 25 produce a different distance tomorrow?

Battery condition, surface friction, wheel alignment or payload may change; the command is not a guaranteed physical speed.

Misconception to address

Equal speed commands do not guarantee perfectly equal wheel motion.

Troubleshooting

If motors do not run, check robot batteries and switch; USB alone is not the rover drive supply. If direction is wrong, stop and check the correct extension and program.

Stretch task

Compare two low speed settings with three trials each. Use mean distance to estimate the difference.

Design a cargo cradle

Learning evidence

Build and compare two removable cargo cradles using consistent trials and a stated design constraint.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Parcel stays in place for three short trials; the sensor view, wheels, controls and switch remain unobstructed.

Exit question and answer key

Why should you keep the parcel and code unchanged when comparing the cradles?

It helps isolate the effect of the cradle design; changing several factors would make the cause of a difference unclear.

Misconception to address

The supplied chassis is already built. Bracing a cardboard attachment is different from redesigning the robot’s electronics.

Troubleshooting

If the attachment shifts or interferes, stop and remove it. Reduce size or reposition it; do not compensate by increasing speed.

Stretch task

Have another pair reproduce your attachment from the drawing, then compare results.

Give the rover reliable senses

Learning evidence

Compare distance readings at three measured positions and explain one unreliable reading.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Nine readings recorded; measured and reported distances compared; one limitation explained.

Exit question and answer key

The sensor returns zero. Should your moving robot assume there is no obstacle?

No. Treat a zero or invalid reading as uncertain and stop; test the target position and sensor before continuing.

Misconception to address

A sensor reading is an estimate, not perfect knowledge of the environment.

Troubleshooting

Use a wide, upright flat target in front of the sensor. Check for a blocked sensor or low batteries. Never solve a missing reading by removing the stop rule.

Stretch task

Calculate the mean of each three-reading set and compare it with the ruler value.

Code a delivery route

Learning evidence

Program a two-leg route and use three trials to tune one timing variable.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

A documented two-leg route ends within the team’s stated bay tolerance in at least two of three trials, and stops every time.

Exit question and answer key

Why can a route succeed once but miss the bay on the next run?

Open-loop timing cannot correct for differences in traction, batteries, start direction or load; repeated trials reveal reliability.

Misconception to address

A “turn right” instruction in a grid represents a precise quarter-turn; a real motor needs calibration or feedback.

Troubleshooting

Shorten the sequence and test its legs separately. Check the start direction before changing the code.

Stretch task

Compare a long route with a shorter route. Keep a record of program length and successful trials.

Stop before an obstacle

Learning evidence

Demonstrate a sensor-controlled stop before contact in three trials, then explain the threshold using data.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Three stops before contact, each with a measured gap, plus an explanation of one failure or limitation.

Exit question and answer key

Why might increasing the stopping threshold help a rover that reacts too late?

It starts the stop decision farther from the obstacle, allowing margin for sensing and motor response. It still needs real tests.

Misconception to address

Obstacle detection does not make this a safety-certified device. The teacher retains direct supervision and switch access.

Troubleshooting

If a reading is zero, stop and recheck the sensor view. If the program continues after stopping, check the latched stopped state and that no second routine restarts motors.

Stretch task

Add an on-screen reason for the stop: obstacle, invalid reading, operator stop or time limit.

Plan the delivery mission

Learning evidence

Combine a cargo attachment, a bounded route and a sensor stop into a mission with measurable acceptance criteria.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

At least two of three full mission trials meet all three criteria; every trial remains bounded and supervised.

Exit question and answer key

Your first version passed 1 of 3 trials and your revision passed 2 of 3. What can you claim?

The revision performed better in these small trial sets. More trials are needed before claiming broad reliability; do not claim it always works.

Misconception to address

A visually impressive run is not enough; the criteria must be set before testing.

Troubleshooting

If the combined mission fails, test cargo retention, movement and sensing separately to locate the failure.

Stretch task

Use the optional line-sensor investigation in the teacher guide to design a future navigation improvement; no unsupported claim that the current rover follows a line.

Showcase with evidence

Learning evidence

Present a reproducible demonstration and defend a design choice using measurements and a limitation.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

A reproducible handover, three recorded trials, a two-minute evidence-based explanation and an individual reflection.

Exit question and answer key

What should you say if the showcase demonstration fails?

Describe what happened, stop safely, compare it with the prior trial record and identify a specific next test. Never hide or replace the recorded result.

Misconception to address

Engineering success includes finding and explaining limits, not only achieving a perfect demonstration.

Troubleshooting

Use the saved program and setup drawing to reproduce conditions. If time is short, demonstrate a verified subsystem and report the full mission’s limits honestly.

Stretch task

Propose one additional user requirement, explain its tradeoff and design a test without claiming the feature is already implemented.

Assess the work, not the spectacle

Score each dimension 1-4 (16 points total). Use individual explanations to distinguish understanding within a team. A teacher intervention for a safety issue prompts reteaching before another powered run.

Dimension	4: Independent	3: Secure	2: Developing	1: Beginning
Mission & constraints	Measurable criteria; reproducible safe setup.	Clear criteria; minor setup clarification.	Some criteria vague or omitted.	Criteria or operating plan absent.
Program explanation	Accurate sequence and conditional; explains limits.	Explains input, rule and output correctly.	Partial explanation; needs prompts.	Cannot connect code to behaviour yet.
Testing & iteration	Repeated fair trials; justified revision and limitation.	Records repeated trials; uses evidence.	Incomplete controls or records.	Claims without usable records.
Communication & roles	Peer can reproduce; each learner explains their work.	Clear handover; shared roles.	Handover or participation uneven.	Handover unclear; little individual evidence.

Sample feedback

“Your stop rule is clearly explained. Your three trials changed both speed and threshold, so the evidence does not identify the cause. Keep speed fixed and compare two thresholds with three trials each.”

Optional line-sensor investigation

This is an extension investigation, not a supplied line-following program. With motors stopped, compare each underside sensor over white paper and dark tape. Use Kitronik’s `readSensor(LfSensor.Left/Right)`, record values, and identify a candidate threshold. Check ambient light and sensor height. Design a state table before attempting motor control, retaining operator stop and a time bound.

Mission Lab and portfolio handover

Simulator facilitation

Students choose a map, write a prediction, add commands, step through the result, record a trial and download the CSV log. Turns are exact quarter-turns; forward is exactly one cell. “If blocked” checks the adjacent cell, then turns right when blocked or moves forward when clear. It does not continuously drive around obstacles.

Teacher solution routes

First delivery: F F F R F F F F.

Around the corner: F R F F F R F F.

Sensor decision: F I F F F F.

Explain the portfolio honestly

This sample demonstrates curriculum design, alignment of practice and assessment, original 3D visual production, synthesized narration, and an interactive logic model. Present physical-robot calibration and classroom validation as next steps. This custom web course does not establish Storyline/Rise authoring proficiency by itself; the original course is a Rise export.

Suggested interview walkthrough (4 minutes)

1. Explain the learner and performance goal. 2. Show a build card and its matching evidence log. 3. Demonstrate a failed simulator route and the learner feedback. 4. Show the teacher rubric, narration/transcript and one visual; explain the hardware-testing boundary.

Sources and credits

Original course

<https://luvbuniz.github.io/robotics/>

Kit and assembly resources

<https://kitronik.co.uk/products/5683-kitronik-move-motor-for-the-bbc-micro-bit>

Manufacturer MakeCode extension

<https://makecode.microbit.org/pkg/KitronikLtd/pxt-kitronik-move-motor>

micro:bit safety

<https://microbit.org/get-started/user-guide/electrical-product-guidance/>

MakeCode editor

<https://makecode.microbit.org/>

Manufacturer API reviewed September 2026. API-compatible source does not replace a successful MakeCode compile or hardware test. Narration uses Amy Sullivan’s own Voicebox clone. Rover illustration is an original conceptual Blender model, not a manufacturer CAD file. Course content is newly authored around the original eight-lesson structure.